



Gurinder Singh

Staff Software Engineer (Mobile & AI)

LLM Agents · RAG Pipelines · Mobile · Automation

Portfolio

LinkedIn

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SCAN FOR PORTFOLIO

90M+

USERS IMPACTED

28+

HACKATHONS

2×

1ST PLACE WINS



PATENT HOLDER

SUMMARY

Staff Software iOS/Mobile & Applied AI Engineer with over a decade of experience architecting and scaling production consumer platforms and AI-powered systems used by millions. Patent holder and technical leader with deep expertise in iOS platform architecture, distributed client systems, and AI-enabled product development—focused on translating emerging AI capabilities into reliable, real-world products.

Proven track record of taking ideas from early prototypes to globally deployed products, spanning startup innovation through enterprise-grade systems serving 90M+ users, most recently at PayPal/Venmo. Experienced in building agentic AI systems, RAG pipelines, and LLM-powered workflows alongside large-scale mobile features and strategic partnerships, with a strong product mindset and comfort operating in ambiguous, cross-functional environments.

Brings a human-centered perspective to applied AI, emphasizing the elevation of human skill, judgment, and ethos through careful system design. Actively invested in mentoring engineers, shaping technical culture, volunteering with NYC high schools and nonprofits, and building products grounded in context, nuance, and real-world use at scale.

TECHNICAL SKILLS PORTFOLIO * Condensed overview — full technical depth available on portfolio

CATEGORY	SPECIALIZED DOMAINS	TOOLS & TECHNOLOGIES	KEY TECH
Mobile	iOS Core Spatial & Hardware Cross-Platform	Swift, Objective-C, SwiftUI, UIKit, SwiftData, Core Data, ARKit, RoomPlan, BLE, NFC, MapKit, CoreML, MLX, RxSwift, Combine, Flutter & Dart	
AI/ML	Models & LLMs Orchestration Media & Vision	Python, PyTorch, OpenAI, Anthropic, Gemini, Llama LangChain, LlamaIndex, CrewAI, N8N, MCP Servers Stable Diffusion, YOLO, OpenCV, Whisper, ElevenLabs	
Cloud & DevOps	Infrastructure BaaS & CI/CD	AWS (Lambda/EC2), GCP (Vertex AI), Azure, Docker Supabase, Firebase, GitHub Actions, Jenkins, Fastlane	
Data & Tools	Databases API & Search Workflow	PostgreSQL, MongoDB, Pinecone, ChromaDB, Redis, GraphQL, Figma, Jira, TDD, Agile, Cursor, Xcode	

PROFESSIONAL EXPERIENCE



PayPal / Venmo

2020 – Present

Staff iOS Engineer → Applied AI/ML Engineer

AI/ML Evolution & Technical Innovation (2023–Present)

Project Mercury — AI-Powered Agentic E-Commerce Platform

June 2025 → Present

- Built fully functional prototype in 72 hours; presented to SVP of AI, earning immediate backing and leading a high-velocity team
- Trained five SDXL LoRA models achieving **23% loss reduction** and 109MB optimized models for production deployment

Agent0 — Autonomous AI Agent for Engineering Teams

February 2025

- Deployed Agent Zero (agent0), an open-source agentic AI framework, to Venmo and PayPal engineering teams — enabling autonomous task execution, intelligent tool use, and self-correcting workflows across development pipelines

AI/ML Implementation & Automation (2024)

- Developed AI-powered crash detection system using RAG pipelines + Vector stores with automated DataDog monitors — **now adopted team-wide**
- Created **10%+ productivity improvements** via AI-powered automation tools
- Championed AI coding assistants across teams, training colleagues on leveraging AI

iOS Engineering Excellence & Business Impact

Major Product Launches & Partnerships

- **Pay With Venmo Checkout (2024–Present):** Reduced end-to-end pay sheet render latency by ~2 seconds
- **Venmo Gift Cards (2023):** Led iOS architecture and delivery for Venmo Gift Cards, supporting major merchants like Amazon and Starbucks
- **Venmo × Amazon (2022):** Built sandbox/integration app in two days for partnership launch
- **QRC Touch-Free Payments (2021):** Led moonshot initiative winning **\$60M CVS contract**
- **Amex Partnership (2020):** Built payment-splitting integration for **\$400M partnership deal**

iOS Innovation & Patents

- **QRC Widget Patent (2022):** Filed "Interface Widget Tool for Automatic QR Code Generation" (PayPal Ref. OCP.D2021.103924.US1). First team to use SwiftUI for iOS widgets at Venmo. **Reduced user friction by 66%**
- **Dynamic Island / Live Activities / App Clip:** Built end-to-end infrastructure including HLDs, LLDs, and APNS notification systems
- **Visual Snapshot Testing:** Built comprehensive visual snapshot testing suite using XCTest for UI regression testing

Technical Leadership & Impact

- Partnered with Design Systems to create modern SwiftUI component library that **cut feature development time by 30%**



Google Stadia

2019 – 2020

iOS Developer/Consultant

- Worked on Google Stadia for mobile platforms using Flutter/Dart and native iOS plugins
- Implemented Core Bluetooth integration for Stadia controller connectivity and low-latency communication
- Utilized Xcode Instruments for performance profiling and CPU/GPU optimization



Morgan Stanley

2018 – 2019

iOS Developer/Consultant

- Worked on Hybrid (Cordova + Native) Banking and Wealth Management apps for iOS

- Developed key features: check-deposit, biometric authentication, certificate-pinning, jailbreak detection, push authentication
- Migrated codebase from 98% Objective-C to 80% Objective-C / 20% Swift

HACKATHON ACHIEVEMENTS & INNOVATION * Featured highlights — full hackathon history on portfolio

28+ hackathons attended across **6+ countries** as participant, mentor, and judge. Earned **2 first-place wins** and served as judge at Yale (2×). Recognized globally for innovation and technical excellence.

 **1ST PLACE**



MLH @ Bloomberg London (2017)
 Automated Search & Rescue System.
 Featured in BBC documentary.

 **1ST PLACE**



Devcamp NYC (2016)
 Talent Management App. Won among 300+ participants at Google NYC.

 **FINALIST**



HackZurich (2018)
 AR Car Buying Tool. Finalist among 400+ participants.



Hack Yale – Judge (2×)
 Yale University



PenApps – Mentor
 University of Pennsylvania

SELECT PERSONAL PROJECTS * Curated highlights — explore 20+ projects on portfolio



FEATURED

Ethereal Dimension
 AI/ML & Spatial Computing
Austin DoT Partnership:
 Built CV system for real-time barricade detection & compliance.
Hylios:
 AR room scanner using RoomPlan/ARKit for 3D spatial intelligence. Available on App Store.

Computer Vision YOLO ARKit
 RoomPlan CoreML



FEATURED

VectorVerse Evolve
 AI Automation & SEO
Content Pipeline:
 Fully automated social media platform with n8n workflows & Strapi CMS. 3-stage pipeline: Intelligence → Ideation → Generation.
Agentic SEO:
 AI-powered SEO strategist with keyword research, geographic targeting & magic reports.
Open-WebUI:
 Built custom deployment for Azure AI Foundry integration.

n8n Strapi Next.js Azure AI
 Open-WebUI



FEATURED

Rosicrucian Knowledge Explorer
 AI Search & Media Platform
 Full-stack RAG search engine for 73K+ vectors from digests, books, symposiums, podcasts, YouTube videos.
 PDF viewer with page-jump & highlighting, video and podcast player with timestamp seeking. Qdrant + OpenAI + R2. 60-test suite. Amplitude analytics.

Next.js Qdrant R2 OpenAI
 Supabase

EDUCATION



BS in Computer Systems (Deferred)
 New York City College of Technology, Brooklyn

2012 – 2016

LEADERSHIP & COMMUNITY

-  **Advisory Board Member** for multiple tech startups on iOS architecture and AI/ML implementation
-  **Interfaith ERG Chapter Lead (Sikh community):** Organized events, secured speakers, created [Gurdwara Langar Seva](#) game to teach Sikh Seva Traditions